

Biology, 2018 EC, entrance exam, version 7

Grade 12 · Biology · 2018 E.C.

Time allowed: 120 minutes

Questions: 237

Total marks: 237

Name _____ Grade _____ ID No. _____ Date _____

1. The virus that uses a bacterium to replicate its genetic material is called (1)
 A. Adenovirus B. Bacteriophage
 C. H1N1 virus D. Herpes simplex

2. The device for injecting cells with a desired genetic information is (1)
 A. transduction B. PCR
 C. translocation D. gene gun

3. The mode of nutrition in fungi cannot be (1)
 A. heterotrophic B. saprobiotic
 C. autotrophic D. parasitic

4. Nutrients locked up in organic molecules are released to the environment by (1)
 A. denitrifiers B. decomposers
 C. mycorrhizae D. herbivores

5. With nitrogen, organisms cannot synthesize one of the following which one? (1)
 A. Adenosine triphosphate B. Glycolipid
 C. Deoxyribonucleic acid D. Ribonucleic acid

6. What do you call the transfer of genetic material from one microorganism to another by viral agent? (1)
 A. Transamination B. Transduction
 C. Translocation D. Transformation

7. What is the composition of the cell wall of prokaryotic cells? (1)
 A. Pectin B. Cellulose
 C. Protein D. Peptidoglycan

8. The protein part of a holoenzyme is called (1)
 A. Coenzyme B. apoenzyme
 C. cofactor D. Prosthetic group

9. Inhibitors that are often similar in shape to the substrate and fool the enzyme is (1)
 A. an activator B. a competitive inhibitor
 C. a non-competitive inhibitor D. an irreversible inhibitor

10. Which one of the following is not a reducing sugar? (1)
 A. maltose B. Lactose
 C. Galactose D. Sucrose

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- 11.** Who is Dr. Akliu Lemma and what is his major contribution to biological research? (1)
- A. A biologist who developed a vaccine for HIV/AIDS
 - B. A biologist who studied the impact of chemical pesticides on humans
 - C. A biologist who used Endod plants to control schistosomiasis
 - D. A researcher who discovered ancient human fossils
-
- 12.** What is the main focus of the research at the Armauer Hansen Research Institute (AHRI)? (1)
- A. Agricultural research and plant breeding
 - B. The study of human diseases like malaria, tuberculosis, and HIV/AIDS
 - C. Development of vaccines for cattle diseases
 - D. Conservation of Ethiopian biodiversity
-
- 13.** Which institution in Ethiopia is dedicated to the preservation of genetic diversity in plants? (1)
- A. Ethiopian Health and Nutrition Research Institute (EHNRI)
 - B. Institute of Biodiversity Conservation (IBC)
 - C. Addis Ababa University (AAU)
 - D. Akliu Lemma Institute of Pathobiology (ALiPB)
-
- 14.** What did Professor Tilahun Yilma contribute to the field of biology? (1)
- A. Development of a vaccine for rinderpest in cattle
 - B. Discovery of ancient human fossils in Ethiopia
 - C. Research on the impact of pesticides on plants
 - D. Studies on human nutrition and health
-
- 15.** What is the role of the Ethiopian Institute of Agricultural Research (EIAR)? (1)
- A. To study human diseases and develop cures
 - B. To develop agricultural techniques for improved farming practices
 - C. To conserve Ethiopia's biodiversity, including plant and animal species
 - D. To conduct research in human nutrition and health
-
- 16.** Which biologist is known for discovering a 3.3 million-year-old humanoid child fossil in 2006? (1)
- | | |
|---------------------------------------|----------------------------|
| A. Dr. Tewolde Berhan Gebre Egziabher | B. Dr. Zeresenay Alemseged |
| C. Professor Tilahun Yilma | D. Dr. Gebissa Ejeta |
-
- 17.** What is the primary research focus of the Akllu Lemma Institute of Pathobiology (ALiPB)? (1)
- A. Development of plant-based vaccines
 - B. Research on the prevention and treatment of animal diseases
 - C. Microbiology, disease vectors, and parasitic diseases
 - D. Human nutrition and food security
-
- 18.** In which area is Dr. Tewolde Berhan Gebre Egziabher best known for his work? (1)
- A. Developing a vaccine for HIV/AIDS
 - B. Securing community rights and advocating for biosafety and biodiversity
 - C. Discovering ancient human fossils
 - D. Conducting agricultural research on drought-resistant crops

19. Which university is home to the Akllu Lemma Institute of Pathobiology (ALiPB)? (1)

- A. Haramaya University** **B. Mekelle University**
C. Addis Ababa University **D. Jimma University**

20. Which of the following is NOT one of the research areas at the Ethiopian Health and Nutrition Research Institute (EHNRI)?

- A. HIV/AIDS research
- B. Tuberculosis diagnosis and treatment
- C. Genetic research on domestic plants
- D. Health and nutrition issues affecting public health

21. Which of the following is part or branch of biotechnology? (1)

- A. Physiology** **B. Immunology**
C. Gene technology **D. Parasitology**

22. Which of the following microorganisms has no double membrane? (1)

- A. Bacteria
B. Protozoa
C. Yeast
D. Algae

23. Which of the following do we use genetically manipulated bacteria for purpose other than human health? (1)

- A.** Production of insulin **B.** Production of interferons
C. Production of pesticides **D.** Production of vaccines

24. Which of the following groups of viruses copy their RNA to DNA using reverse transcriptase enzyme? (1)

- A.** Bacteriophages
B. RNA viruses
C. Retroviruses
D. DNA viruses

25. What is the reason that HIV targets CD4 T-lymphocytes? Because T-lymphocytes are: (1)

- A. coordinators of the immune system
B. reduced in number than any other cells
C. weak to fight HIV
D. found at the CD4 cells

26. Why do microorganisms decompose dead organic matter? (1)

- A.** To drive energy for their own uses **B.** To keep the balance of matter
- C.** To release minerals for plants **D.** To clean up the environment

27. Which of the following microorganism use hyphae to release extracellular enzymes to obtain their nutrition? (1)

- A. Bacteria** **B. Viruses**
C. Fungi **D. Algae**

28. The first antibiotic penicillin was produced from (1)

- A. fungus
B. viruses
C. bacteria
D. protozoans

29. What is the nutritional mode of organisms that live in close association of other organisms for mutual benefit? (1)

- A.** Saprobiotic **B.** Parasitic
C. Autotrophic **D.** Symbiotic

30. In which organ of the human body is more normal microbiota found? (1)

- A.** Kidneys
B. Small intestine
C. Large intestine
D. Stomach

31. Which of the following is NOT an infectious disease? (1)

- A.** Tuberculosis
- B.** Coronary heart disease
- C.** AIDS
- D.** The common cold

32. Which of the following is effective to reverse enzyme inhibition that is caused by a reversible competitive inhibition? (1)

- A.** The pH of the reaction
- B.** The temperature of the reaction
- C.** The substrate concentration
- D.** The concentration of the inhibitor

33. What do we call substances that bind with an enzyme away from the active sites and cause a change in the structure of the active site? (1)

- A. Cofactors** **B. Allosteric effectors**
C. Coenzymes **D. Substrate**

34. Which of the following is the common characteristic of all enzymes? (1)

- A.** They act inside cells only.
- B.** They operate at any pH.
- C.** They lower activation energy.
- D.** They require cofactors.

35. Which statement is true about enzymes? Enzymes (1)

- A.** can be reused **B.** change after the reactions
C. are lipids **D.** increase activation energy

36. The cofactors that are found in the enzymes include (1)

- A.** proteins and lipids **B.** vitamins and amino acids
C. sugars and iron **D.** riboflavin and zinc

37. Which class of enzymes joins molecules together? (1)

- A. Lyases B. Hydrolases
C. Transferases D. Ligases

38. The protein part of an enzyme is called a(an) (1)

- A. coenzyme
B. apoenzyme
C. cofactor
D. holoenzyme

39. The rate of chemical reaction is increased by (1)

- A. decreasing the activation energy
B. cooling the reaction
C. decreasing the concentration of reactants
D. removing a catalyst

-
- 40.** Which of the following statements is true about the lock and key enzyme action? (1)
- A. The active site and substrate are not complementary in shape
 - B. The shape of the substrate and the active site are complementary
 - C. The binding of the substrate produces conformational change
 - D. The model offers a better explanation to enzymes than induced fit model
-
- 41.** Vectors that are used to transport foreign genes into a bacterial host are (1)
- A. viruses
 - B. agrobacteria
 - C. plasmids
 - D. gene gun
-
- 42.** To which group of viruses do HIV belong? (1)
- A. Bacteriophages
 - B. RNA viruses
 - C. Retroviruses
 - D. DNA viruses
-
- 43.** All of the following statements are correct, EXCEPT (1)
- A. All viruses are parasites
 - B. Gram negative bacteria are highly susceptible to antibiotics
 - C. Endotoxins are structural components of Gram negative bacteria
 - D. Gram positive bacteria have thick peptidoglycan
-
- 44.** Which of the following makes T-helper cells more vulnerable to HIV infection? (1)
- A. The presence of gp120 on their surface
 - B. The presence of HIV receptors on their surface
 - C. The absence of antibodies on their surface
 - D. The presence of an envelope on their surface
-
- 45.** Which of the following is TRUE about viruses? (1)
- A. Viruses have independent reproduction
 - B. Only some viruses are parasites
 - C. Viruses have cellular organelles
 - D. Viruses have genetic material
-
- 46.** In nitrogen cycle, plants use nitrogen in the form of (1)
- A. nitrites
 - B. ammonium
 - C. nitrates
 - D. nitrogen
-
- 47.** What is the difference between chromosome and gene? (1)
- A. Gene is a threadlike structure
 - B. Genes are found on the chromosomes
 - C. Chromosome is found on the gene
 - D. Both genes and chromosomes are the same
-
- 48.** A group of related genera is known as (1)
- A. species
 - B. order
 - C. family
 - D. genus
-
- 49.** In a cross between round green pea of RRyy genotype and wrinkled-yellow rryy genotype, what is the expected genotype of the F1? (Note: The given options have typos; likely rryy is rrYY for wrinkled yellow). The correct cross: RRyy (round green) × rrYY (wrinkled yellow) → all RrYy (round yellow). The options are: A.RRyy, B.rry, C.RRyy, D.RrYy. (1)
- A. RRyy
 - B. rry
 - C. RRyy
 - D. RrYy

-
- 50.** Which of the following is known as code of life? (1)
- A. DNA
B. Chromosome
C. Nucleotide
D. Nucleoside
-
- 51.** A type of dominance where the expression of one allele masked by another allele is called (1)
- A. complete dominance
B. incomplete dominance
C. co dominance
D. independent dominance
-
- 52.** The phenotypic ratio of mendelian inheritance in dihybrid cross of F₂ generation is (1)
- A. 9:3:3:1
B. 9:3:3:2
C. 3:1
D. 2:2:1
-
- 53.** Why do we call meiosis is reduction division? Because (1)
- A. it results diploid cells
B. it is the result of genetic variation
C. it results haploid cells
D. crossing over takes place
-
- 54.** A plant has a genotype Aa Bb. There is no linkage of the genes. The gametes it will produce are (1)
- A. AB and ab
B. Aa and Bb
C. AA, aa, BB, and Bb
D. AB, Ab, aB and ab
-
- 55.** In which stage of meiosis do pairs of homologous chromosomes split up? (1)
- A. Prophase
B. Anaphase
C. Metaphase
D. Telophase
-
- 56.** How many chromosomes do humans inherit from each of their parents? (1)
- A. 23
B. 46
C. 23 pairs
D. 22 pairs
-
- 57.** What are bacteriophages? (1)
- A. Bacteria infecting viruses
B. Bacteria that used for genetic engineering
C. Plasmid vectors that used to carry genes
D. Enzymes used to cut DNA
-
- 58.** Which of the following microorganisms include both multicellular and unicellular organisms? (1)
- A. Bacteria
B. Paramecium
C. Amoeba
D. Fungi
-
- 59.** Cholera and typhoid can be transmitted by (1)
- A. viruses
B. fungi
C. protozoan
D. bacteria
-
- 60.** The protein shell of the virus is called (1)
- A. nucleic acid
B. genetic material
C. nucleotide
D. capsid

61. Which element is released from dead matter as hydrogen sulphide giving rotten eggs smell? (1)

- A. Hydrogen
B. Nitrogen
C. Carbon
D. Sulphur

62. Which of following process add carbon-dioxide to the atmosphere? (1)

- A. Fossilization**
C. Photosynthesis
- B. Respiration and photosynthesis**
D. Respiration

63. What we call organisms that feed on primary consumers? (1)

- A. 1st trophic level
B. 2nd trophic level
C. 3rd trophic level
D. 4th trophic level

64. What would be the possible genotype ratio of calves if two roans (Rr) are mated? (1)

- A. 2 RR : 1 Rr : 1 rr
B. 1 RR : 2 Rr : 1 rr
C. 1 RR : 2 Rr : 1 r
D. 0 RR : 2 Rr : 2 r

65. What is the term that better describes the genetic makeup of an organism? (1)

- A.** Gene **B.** Allele
C. Genotype **D.** Phenotype

66. Which group of organic compounds include enzymes? (1)

- A.** Proteins
- B.** Lipids
- C.** Starches
- D.** Carbohydrates

67. In competitive inhibition, which one of the following factors determines the rate of Inhibition? (1)

- A. Temperature
B. Enzyme concentration
C. Substrate concentration
D. Ratio of inhibitor to enzyme concentration

68. Which type of enzymes are used by decomposers during saprobial nutrition? (1)

- A.** Hydrolytic enzymes **B.** Ligase enzymes
C. Protease enzymes **D.** RNase enzymes

69. Which blood group are codominant? (1)

- A. IA and i B. IB and i
C. IA and IB D. IA and IAB

70. Which bacteria converts nitrates to atmospheric nitrogen? (1)

- A. Nitrifying bacteria
B. Denitrifying bacteria
C. Nitrogen fixing bacteria
D. Ammonifying bacteria

71. Which group of micro-organisms causes the diseases known as athlete's foot? (1)

- A. Fungi
B. Bacteria
C. Viruses
D. Protozoa

-
- 72.** Which of the following is different from the others? (1)
- | | |
|--|--|
| A. Genetically modified organisms | B. Genetically engineered organisms |
| C. Pathogenic organisms | D. Transgenic organisms |
-
- 73.** In report of scientific experiment, description of which of the following comes before all the others? (1)
- | | |
|----------------------|----------------------|
| A. Procedure | B. Results |
| C. Hypothesis | D. Conclusion |
-
- 74.** Which of the following bases is not found in RNA? (1)
- | | |
|--------------------|-------------------|
| A. Adenine | B. Guanine |
| C. Cytosine | D. Thymine |
-
- 75.** Which level of structural organization involves two or more polypeptide chains? (1)
- | | |
|------------------------------|--------------------------------|
| A. Primary structure | B. Secondary structure |
| C. Tertiary structure | D. Quaternary structure |
-
- 76.** Among the following factors that affect enzyme activity, which one does not cause enzyme denaturation? (1)
- | | |
|----------------------------|---------------------------------|
| A. High pH | B. Low pH |
| C. High temperature | D. Competitive inhibitor |
-
- 77.** Most co-enzymes are the derivatives of which of the following nutrients? (1)
- | | |
|-------------------------|--------------------|
| A. Proteins | B. Vitamins |
| C. Carbohydrates | D. Fats |
-
- 78.** To produce glucose from starch which of the following enzymes would one use? (1)
- | | |
|----------------------|-------------------|
| A. Isomerase | B. Lyases |
| C. Hydrolases | D. Ligases |
-
- 79.** Which of the following substances can be used over and over again in an enzyme catalyzed reaction? (1)
- | | |
|------------------------------------|----------------------------|
| A. Enzyme substrate complex | B. Enzyme |
| C. Substrate | D. Reaction product |
-
- 80.** What is an apo enzyme? (1)
- | | |
|----------------------------------|----------------------------------|
| A. protein molecule | B. An active enzyme |
| C. A non-protein molecule | D. A non-protein cofactor |
-
- 81.** What is the intermediate compound that is formed during an enzyme catalyzed reaction? (1)
- | | |
|---------------------|------------------------------------|
| A. Reactant | B. Product |
| C. Substrate | D. Enzyme substrate complex |
-
- 82.** Which of the following stages of mitosis cell division does the sister chromatids aligned at equator of cell plate? (1)
- | | |
|----------------------------------|---------------------------|
| A. Interphase of prophase | B. End of prophase |
| C. Metaphase | D. Telophase |
-
- 83.** Which of the following stages of mitosis do chromatids pulled to opposite poles of the cell? (1)
- | | |
|--------------------|---------------------|
| A. Anaphase | B. Metaphase |
| C. Prophase | D. Telophase |

84. What happens to an animal cell placed in a hypertonic solution? (1)

- A.** It swells and bursts.
B. It shrinks due to water loss.
C. It remains unchanged.
D. It becomes turgid.

85. Which of the following is an example of an isotonic solution? (1)

- A.** Pure water
- B.** 5% salt solution
- C.** Blood plasma
- D.** 10% sugar solution

86. The movement of water molecules across a selectively permeable membrane from a region of high concentration to a region of low concentration is called (1)

- A. diffusion
B. osmosis
C. active transport
D. facilitated diffusion

87. Which transport process requires energy (ATP)? (1)

- A.** Simple diffusion **B.** Osmosis
C. Active transport **D.** Facilitated diffusion

88. A plant cell in a hypotonic solution will (1)

- A. undergo plasmolysis
B. become turgid
C. shrink
D. burst

89. Which type of microscope provides the highest magnification and resolution? (1)

- A. Light microscope
B. Transmission electron microscope
C. Scanning electron microscope
D. Compound microscope

90. The total magnification of an image viewed under a 10x eyepiece and a 40x objective lens is: (1)

- A. $400x$
B. $50x$
C. $100x$
D. $40x$

91. The part of the microscope that controls the amount of light passing through the specimen is the: (1)

- A. condenser
B. objective lens
C. diaphragm
D. coarse focus knob

92. Which microscope is best for viewing the surface structure of a specimen? (1)

- A.** Compound light microscope **B.** Scanning electron microscope
C. Transmission electron microscope **D.** Phase-contrast microscope

93. Which of the following is a function of the fine adjustment knob on a microscope? (1)

- A. Moves the stage up and down for coarse focusing
- B. Increases magnification
- C. Sharpens the image at higher magnifications
- D. Controls the light intensity

94. Who is credited with coining the term 'cell' after observing cork under a microscope? (1)

- A.** Anton van Leeuwenhoek **B.** Robert Hooke
C. Matthias Schleiden **D.** Theodor Schwann

-
- 95.** The cell theory states that (1)
- A. all organisms are made of multiple cells
 - B. cells arise only from pre-existing cells
 - C. only plant cells have a nucleus
 - D. cells can be created from non-living material
-
- 96.** Which of the following structures is found in all types of cells? (1)
- A. Nucleus
 - B. Mitochondria
 - C. Cell membrane
 - D. Chloroplast
-
- 97.** Which scientist is known for observing living microorganisms with a simple microscope? (1)
- A. Rudolf Virchow
 - B. Matthias Schleiden
 - C. Anton van Leeuwenhoek
 - D. Theodor Schwann
-
- 98.** Prokaryotic cells differ from eukaryotic cells in that they (1)
- A. lack a cell membrane
 - B. do not have a nucleus
 - C. have membrane-bound organelles
 - D. are larger and more complex
-
- 99.** Which type of microscope is used to directly observe living plant & animal cells and their components? (1)
- A. Transmission electron microscope
 - B. Scanning electron microscope
 - C. Electron microscope
 - D. Light microscope
-
- 100.** If a cell fails to make an enzyme, which one of its organelle is most likely not functioning? (1)
- A. Mitochondria
 - B. cell membrane
 - C. Ribosomes
 - D. Vacuole
-
- 101.** The chemical digestion of proteins begins in the (1)
- A. mouth
 - B. stomach
 - C. small intestine
 - D. large intestine
-
- 102.** The breathing movements are controlled by two different sets of muscles that change the pressure in the chest cavity. Which one of the following happens when we breathe in? (1)
- A. Ribs move down and inwards
 - B. Diaphragm relaxes and becomes dome-shaped
 - C. The pressure inside the chest decreases
 - D. The volume of the chest decreases
-
- 103.** Where does the exchange of gases take place in the human respiratory system? (1)
- A. Nasal passage
 - B. Trachea
 - C. Alveoli
 - D. Bronchiole
-
- 104.** A student performs a blood group test and finds that the red blood cells of an individual possess both A and B antigens, and blood plasma contains no anti-A or anti-B antibodies. Which of the following statements is not true about this person? (1)
- A. His blood group is AB
 - B. He can receive blood from individuals with blood group AB
 - C. He can receive blood from individuals with blood group A
 - D. His blood group is O

-
- 105.** Where does the transfer of oxygen from blood to the body cells take place? (1)
- A. Arteries
B. Arterioles
C. Capillaries
D. Venules
-
- 106.** Identify the incorrect statement regarding the structural and functional differences between arteries, veins, and capillaries. (1)
- A. Arteries have thicker walls than veins
B. Veins generally have a wider lumen (internal diameter) than arteries
C. Capillaries have thinner walls than both arteries and veins
D. Arteries have valves, while veins have pulse
-
- 107.** Which of the following characteristics does not apply to organisms in the kingdom Fungi? (1)
- A. Saprophytic nutrition
B. Parasitic nutrition
C. Prokaryotic cell structure
D. Eukaryotic cell structure
-
- 108.** Which one of the following phylum is known for having jointed limbs? (1)
- A. Phylum Arthropoda
B. Phylum Echinodermata
C. Phylum Annelida
D. Phylum Coelenterata
-
- 109.** Which of the following plant groups is characterized by a well differentiated body, specialized vascular tissues for transport but lacks seeds and fruits? (1)
- A. Pteridophyte
B. Bryophyte
C. Angiosperm
D. Gymnosperm
-
- 110.** In which types of asexual reproduction do unfertilized eggs develop into new offspring, as in some insects and vertebrates? (1)
- A. Budding
B. Parthenogenesis
C. Fragmentation
D. Binary fission
-
- 111.** In a biology laboratory, you are characterizing an enzyme. Which of the following describes a physical property of the enzyme? (1)
- A. Enzyme reversibility
B. Enzyme regulation
C. Enzyme sensitivity
D. Enzyme precipitation
-
- 112.** What is the term for a group of interacting individuals of the same species living in a particular area? (1)
- A. Community
B. Population
C. Biosphere
D. Ecosystem
-
- 113.** Which of the following microorganisms is primarily responsible for the final stage (methanogenesis) of biogas production? (1)
- A. Archaea
B. Bacteria
C. Fungi
D. Protozoa

-
- 114.** Which of the following correctly identifies the carbon and electron sources utilized by chemolithoautotrophic organisms? (1)
- A. Carbon dioxide and organic compounds, respectively
 - B. Carbon dioxide and inorganic compounds, respectively
 - C. Organic carbon and inorganic compounds, respectively
 - D. Organic carbon and carbon dioxide, respectively
-
- 115.** Identify the statement that is not a defining characteristic of archaea. (1)
- A. Their cell walls contain true peptidoglycan.
 - B. They are prokaryotic organisms.
 - C. They exhibit the capacity to inhabit extreme environmental conditions.
 - D. They reproduce through binary fission, budding, or fragmentation.
-
- 116.** Which member of phylum of fungi establishes a successful symbiotic relationship with the roots of trees? (1)
- | | |
|------------------|------------------|
| A. Ascomycota | B. Zygomycota |
| C. Basidiomycota | D. Glomeromycota |
-
- 117.** Which one of the following disease is not correctly matched with its causative agent? (1)
- A. Tuberculosis → *Mycobacterium tuberculosis*
 - B. Gastroenteritis → Rota viruses
 - C. Cholera → *Vibrio cholera*
 - D. Chancroid → *Treponema pallidum*
-
- 118.** What is the field of Ethiopian biologist Tewolde Berhan Gebre Egzabher? (1)
- | | |
|-------------------|-------------------------|
| A. Marine biology | B. Conservation biology |
| C. Genetics | D. Microbiology |
-
- 119.** What is the purpose of using antiseptics in controlling microorganisms? (1)
- A. To inhibit microbial growth in food products
 - B. To sterilize equipment and media
 - C. To control microorganisms on inanimate surfaces
 - D. To control microorganisms on living tissue
-
- 120.** What is the foremost goal of gene therapy in modern biotechnology? (1)
- A. To cure genetic diseases by replacing defective gene
 - B. To create genetically modified organism
 - C. To produce vaccine against infectious disease
 - D. To produce cloned organism
-
- 121.** What is the role of enzymes in traditional biotechnology? (1)
- A. To encode specific trait in an organism
 - B. To catalyze biochemical reactions in fermentation
 - C. To modify gene of an organism
 - D. To cut DNA into specific gene

-
- 122.** From the following which one is a biotic component of the ecosystem? (1)
- | | |
|-------------|----------------|
| A. Parasite | B. Competition |
| C. Disease | D. Oxygen |
-
- 123.** Which of the following term is used for fungal infection of the nails? (1)
- | | |
|----------------|------------------|
| A. Tineabarbae | B. Tineangum |
| C. Tineapedia | D. Tineacorporis |
-
- 124.** Which of the following bacteria produces acetic acid and does not cause food poisoning? (1)
- | | |
|--------------------------|-------------------------|
| A. Staphylococcus aureus | B. Acetobacter aceti |
| C. Bacillus cereus | D. Campylobacter jejuni |
-
- 125.** Which of the following consists of only spirit products? (1)
- | | |
|----------------------------------|---------------------------------|
| A. Whisky, gin, brandy and vodka | B. Gin, brandy, beer and whisky |
| C. Wine, brandy, gin and vodka | D. Wine, beer, gin and vodka |
-
- 126.** Which of the following organism add flavor by fermenting meats such a salami, summer sausage & Lebanon bologna? (1)
- | | |
|-----------------------------|--------------------------------|
| A. Saccharomyces cerevisiae | B. Streptococcus cremoris |
| C. Lactobacillus plantarum | D. Methylophilus methylophilus |
-
- 127.** Which of the following organism transmits Africa sleeping sickness Trypanosomiasis? (1)
- | | |
|---------------|-------------|
| A. Sand fly | B. Mosquito |
| C. Tsetse fly | D. Housefly |
-
- 128.** Which group of archaeobacteria are called salt loving bacteria? (1)
- | | |
|-----------------|----------------|
| A. Halophiles | B. Methanogens |
| C. Thermophiles | D. Acidophilus |
-
- 129.** Which of the following protozoan disease causes blindness, mental retardation & multi system organ familiarity? (1)
- | | |
|------------------|------------------|
| A. Toxoplasmosis | B. Leishmaniasis |
| C. Trypanosomes | D. Malaria |
-
- 130.** Which one of the following is not correct about lactic acid fermentation (1)
- A. pyruvate converted to lactic acid
 - B. Oxygen is being used up
 - C. Takes place in muscle cell & bacterial cell
 - D. Reduced NAD generate oxidized NAD
-
- 131.** In which of the following features are eukaryotic cells distinguished from prokaryotic cells? (1)
- | | |
|--------------------------------|--|
| A. They have mitochondria. | B. They have smaller ribosomes. |
| C. Their nuclei lack membrane. | D. They lack membrane bounded organelle. |
-
- 132.** The use of microbe to extract metals from ores is known as (1)
- | | |
|----------------------|------------------------|
| A. Bioremediation | B. Bio hydrometallurgy |
| C. Biogas Production | D. Biofuel production |

-
- 133.** Which of the following is not an industrial use of bacteria? (1)
- | | |
|--|---------------------------------|
| A. Recycling of minerals through ecosystems | B. Food processing |
| C. Genetic engineering | D. Waste water treatment |
-
- 134.** Preparation of specimen to see under the microscope is known as (1)
- | | |
|----------------------|-------------------------|
| A. Resolution | B. Mounting |
| C. Staining | D. Magnification |
-
- 135.** Cells that are adapted to carry out a particular body function are (1)
- | | |
|----------------------------------|-------------------------------|
| A. Embryonic stem cells | B. Specialized cells |
| C. Undifferentiated cells | D. Unspecialized cells |
-
- 136.** A cell with a large number of mitochondria is expected to (1)
- | | |
|--|---|
| A. produce more digestive enzymes | B. perform a lot of photosynthesis |
| C. store more food | D. make a lot of ATP |
-
- 137.** Which of the following structures of the cell are involved in the synthesis of protein? (1)
- | | |
|-----------------------|--------------------|
| A. Chloroplast | B. Ribosome |
| C. Golgi body | D. Vacuole |
-
- 138.** Which of the following organelle is the site of photosynthesis? (1)
- | | |
|-----------------------|----------------------|
| A. Lysosome | B. Centriole |
| C. Chloroplast | D. Leucoplast |
-
- 139.** The movement of water molecules from a region of high concentration to low concentration is (1)
- | | |
|---------------------|-----------------------------|
| A. Osmosis | B. Passive transport |
| C. Diffusion | D. Active transport |
-
- 140.** Which of the following is true about rate of diffusion? (1)
- A.** Diffusion increases when the temperature decreases.
- B.** Diffusion increases when the temperature increases.
- C.** Diffusion decreases when there is very high concentration difference between two areas.
- D.** Diffusion increases when there is very low concentration difference between two areas.
-
- 141.** In grade 9 laboratory class, the teacher gives the students a sampler of food and asked them to investigate the presence of glucose. What kind of test and result will enable the students to confirm the presence of glucose respectively? (1)
- | | |
|--|---|
| A. Benedict's test and orange colour | B. Biuret test and purple colour |
| C. Emulsion test and milky white colour | D. Iodine test and blue black colour |
-
- 142.** In a Benedict's test for sucrose, why does the test sample first boil with hydrochloric acid? (1)
- | | |
|---|--|
| A. To reduce Benedict's solution | B. To neutralize the solution |
| C. To hydrolyze non-reducing sugar | D. To retest mixture with Benedict solution |
-
- 143.** Which deficient nutrients are correctly paired with their deficiency diseases? (1)
- | | |
|----------------------------|---------------------------------------|
| A. Protein → Scurvy | B. Carbohydrate → Beri-beri |
| C. Iron → Marasmus | D. Vitamin A → Night blindness |

-
- 144.** When you breathe out during gas exchange, all of the following happens to the breathing movement except? (1)
- | | |
|---|--|
| A. The volume of the thorax decrease | B. Pressure inside chest cavity decreases |
| C. Diaphragm moves upward | D. Pressure inside chest cavity increases |
-
- 145.** The muscle that separate the chest cavity from the abdominal cavity is known as (1)
- | | |
|------------------------|----------------------|
| A. Nasal cavity | B. Trachoma |
| C. Diaphragm | D. Epiglottis |
-
- 146.** While oxygen is carried by the red blood cells, most digested food and wastes are carried by the (1)
- | | |
|---------------------|-----------------------------|
| A. Plasma | B. White blood cells |
| C. Platelets | D. Red blood cells |
-
- 147.** The absolute maximum amount of air you can take into or breathe out of your lungs is called (1)
- | | |
|---------------------------|-----------------------------|
| A. Tidal volume | B. Vital capacity |
| C. Residual volume | D. Breathing hygiene |
-
- 148.** The right side chamber valve which directs the flow of blood by opening and closing is called (1)
- | | |
|--------------------------|---------------------------|
| A. Bicuspid valve | B. Tricuspid valve |
| C. Mitral valve | D. Vena cava |
-
- 149.** Which one of the following statements is true about pulmonary circulation? (1)
- A.** Blood flows from the heart to all around the body and back again.
 - B.** Carry only oxygenated blood from the body
 - C.** Blood flows from the heart to lungs and back again
 - D.** It transport oxygenated blood from the heart and back again
-
- 150.** Which organic molecule in our body has functions as antigens & antibodies of immune system? (1)
- | | |
|--------------------|-------------------------|
| A. Lipids | B. Nucleic acids |
| C. Proteins | D. Carbohydrates |
-
- 151.** The blood group of human being which is commonly referred to as 'universal acceptor' is (1)
- | | |
|--------------------------|-------------------------|
| A. Blood group AB | B. Blood group O |
| C. Blood group A | D. Blood group B |
-
- 152.** All of the following factors that can be increases the risks of developing hypertension except? (1)
- | | |
|--------------------------------|--|
| A. Reducing salt intake | B. Excessive consumption of alcohol |
| C. Increasing age | D. Sedentary lifestyle |
-
- 153.** The relevance and promise of biology as a field of science can be best explained by its application of (1)
- | | |
|--|---|
| A. economic growth and urbanization | B. reducing security problems |
| C. oil and mineral production | D. agriculture and genetic engineering |
-
- 154.** Among the following which one is incorrect statement about the importance of nature for imitating technological products? (1)
- A.** It is the sources of materials that enable imitating
 - B.** Used to develop modern technologies that provide services
 - C.** Used to develop knowledge of problem solving ability
 - D.** To provide limited resources and techniques for society

-
- 155.** Which technological products have been made through imitation of termite mound architecture? (1)
- A. Weaving technology B. Building technology
C. New drone design D. Techniques of preserving vaccine
-
- 156.** The following are the impact of biology on society and natural world except (1)
- A. ensuring food security by increasing productivity
B. lack of involvement in application of microbial action
C. production of antibiotics and treatment for bacterial infection
D. achieving better supply of energy and clean water
-
- 157.** Among which explanation is correctly concerned about ethical issue in biology? (1)
- A. It refers about rightness or wrongness use of biological discoveries
B. Concerned about only ethical treatment of plant species
C. Encourage issues about wrongly uses of technology against human
D. Focus on encouraging unethical treatment of research animals
-
- 158.** Which one of the following is the common characteristic of invertebrate animals? (1)
- A. They have closed circulatory system
B. They have an exoskeleton
C. They are warm-blood animals
D. They possess a backbone and internal skeleton
-
- 159.** Among which of the following organisms reproduce by external fertilization? (1)
- A. Birds B. Rats
C. Crocodiles D. Frogs
-
- 160.** Which mating system of birds those involve with two or more members of the opposite sex during a reproductive season? (1)
- A. Polygyny B. Polygamy
C. Monogamy D. Polyandry
-
- 161.** Which developmental processes of germ layers later develop into organs and organ system? (1)
- A. Organogenesis B. Cleavage
C. Blastula D. Parthenogenesis
-
- 162.** All of the following are considered as a beneficial aspects of insects except? (1)
- A. Effective pollinators B. Biological control agents
C. Sources of useful product and protein D. Non-edible type of animals
-
- 163.** Which one of the following is true about harmful aspect of insect in agriculture? (1)
- A. Damaging crops and reducing production B. Used as pollinators in crop reproduction
C. Production of various medicines D. Production of cosmetics and dyes
-
- 164.** Which one of the following examples of behavior describes when an animal stops responding to stimulus or cue after a period of repeated exposure? (1)
- A. Conditioning B. Reflex
C. Imprinting D. Habituation

- 165.** The Ethiopian wolf lives and hunt in packs. During patrolling their habit, both male and female wolves of the group maintain their area with urine containing pheromone. This pattern of behavior is an example of (1)

A. Operant behavior
B. Territorial behavior
C. Learned behavior
D. Courtship behavior

166. A quick withdrawal of your hand from a hot plate is an example of (1)

A. learned behavior
B. biological clock
C. key stimulus
D. innate behavior

167. Which protein structures have three-dimensional arrangement and unique shape of all amino acids in a single polypeptide chain? (1)

A. Quaternary structure
B. Tertiary structure
C. Primary structure
D. Secondary structure

168. The state in which all people have physical, social and economic access to sufficient, safe and nutritious food is. (1)

A. Food security
B. Nutrition security
C. Nutrition science
D. Balanced diet

169. Recently discovered biosensor can detect all, except (1)

A. Pathogens
B. Herbicides
C. Lactic acid
D. Toxins

170. All are correct about genetic fingerprinting, except (1)

A. It is based on RNA analysis
B. It is unique to individuals except identical twins
C. It is made by gel electrophoresis in the PCR machine
D. It is used for crime investigation

171. Which one of the following is the characteristic of bacteria? (1)

A. Their DNA is contained within the cytoplasm of the cell.
B. They are larger than eukaryotic cell.
C. They have membrane enclosed organelle.
D. They have cell wall made up of lipoprotein.

172. Which one of the following archaea can live in extremely hot temperature? (1)

A. Acidophiles
B. Hyperthermophiles
C. Psychrophiles
D. Psychrotrophs

173. Which one of the following fungi has been used as a test organism to determine Mg, Cu, Zn and Mo in the soil? (1)

A. *Aspergillus flavus*
B. *Claviceps purpurea*
C. *Aspergillus parasiticus*
D. *Aspergillus niger*

174. Which one of the following protozoa causes Chagas disease? (1)

A. *Trypanosoma cruzi*
B. *Trypanosoma brucei*
C. *Toxoplasma gondii*
D. *Plasmodium falciparum*

-
- 175.** All are correct about lysogenic cycle, except (1)
- A. DNA of the virus integrates into the host DNA
 - B. DNA of the virus doesn't integrate into the host DNA
 - C. May or may not kill the host cell
 - D. It takes time
-
- 176.** Which one is the viral symmetry refers to the shape with 20 triangular faces and 12 corners? (1)
- A. Complex
 - B. Helical
 - C. Icosahedral
 - D. Cylindrical
-
- 177.** Which of the following soil bacteria produce powerful protein toxin that used to kill insect pests? (1)
- A. *Metarhizium anisopliae*
 - B. *Beauveria bassiana*
 - C. *Verticillium lecanii*
 - D. *Bacillus thuringiensis*
-
- 178.** All are correct about catabolism, except (1)
- A. It occurs during cellular respiration.
 - B. It is destructive phase of metabolism.
 - C. It is energy releasing reaction.
 - D. It is energy absorbing reaction.
-
- 179.** Where does the light-independent stage of photosynthesis take place? (1)
- A. Grana
 - B. Thylakoids
 - C. Lamina
 - D. Stroma
-
- 180.** Which one of the following photosynthetic pigments found only in the photosynthetic members of Chromista and Dinoflagellates? (1)
- A. Chlorophyll 'a'
 - B. Chlorophyll 'c'
 - C. Carotenoids
 - D. Phycobilins
-
- 181.** All are correct about photosystem II, except (1)
- A. Consists of a pair of Chlorophyll 'a' molecules with an absorption peak at 700nm
 - B. Consists of a pair of Chlorophyll 'a' molecules with an absorption peak at 680 nm
 - C. It passes its electrons to photosystem I
 - D. It replenishes its lost electrons from photolysis of water
-
- 182.** Which one of the following is incorrect about light independent reaction? (1)
- A. It is purely enzymatic
 - B. It is slower than the light reaction
 - C. It is the first step in the mechanism of photosynthesis
 - D. It is the second step in the mechanism of photosynthesis
-
- 183.** Which of the following divisions of Kingdom plantae is correctly matched with its example? (1)
- A. Gymnospermae - true flowering plants
 - B. Pteridophyte - ferns
 - C. Bryophyta - Conifers
 - D. Angiospermae - mosses and liverworts
-
- 184.** Which of the following groups of plants possess roots? (1)
- A. Ferns
 - B. Lichen
 - C. Mosses
 - D. Liverworts

-
- 185.** In which kingdom would you classify the archaea and nitrogen fixing organisms, if the five kingdom system of classification is used? (1)
- A. Plantae
B. Fungi
C. Protista
D. Monera
-
- 186.** Which of the following is not grouped in Protista? (1)
- A. Mucor
B. Giardia lamblia
C. Entamoeba histolytica
D. Diatom
-
- 187.** Representatives of Gymnospermae are (1)
- A. Ferns
B. Conifers
C. Mosses
D. Liverworts
-
- 188.** Which of the following phyla of kingdom animalia contains evolutionary dead organisms? (1)
- A. Coelenterata
B. Nematoda
C. Annelida
D. Porifera
-
- 189.** The smallest/simplest land plants are: (1)
- A. Pteridophytes
B. Bryophytes
C. Angiospermae
D. Filicinophyta
-
- 190.** Which one of the following is the characteristic of Bryophytes? The (1)
- A. absence of vascular tissues
B. presence of reproductive structure in cone
C. presence of well-developed roots, stem and leaves
D. presence of xylem and phloem
-
- 191.** Which of the following is correctly paired with its examples? (1)
- A. Mollusca → segmented worms
B. Nematoda → roundworms
C. Annelida → soft bodied animals
D. Arthropoda → spiny skinned animals
-
- 192.** All are invertebrates, except (1)
- A. Flatworm
B. Cnidaria
C. Coelenterates
D. Chordates
-
- 193.** The most numerous animals in the world are (1)
- A. round worms
B. flatworms
C. mollusks
D. chilopoda
-
- 194.** In which of the following group do blue green algae grouped? (1)
- A. Bryophytes
B. Prokaryotes
C. Protista
D. Fungi
-
- 195.** An important criterion for modern day classification is (1)
- A. resemblances in morphology
B. anatomical and physiological traits
C. breeding habits
D. presence or absence of notochord

-
- 196.** Which of the following organism are found in extreme saline condition? (1)
- | | |
|-------------------------|---------------------------|
| A. Eubacteria | B. Mycobacterium |
| C. Cyanobacteria | D. Archaeobacteria |
-
- 197.** Viruses have (1)
- | | |
|-------------------------------------|-------------------------------|
| A. DNA enclosed in a protein | B. Prokaryotic nucleus |
| C. Single chromosome | D. both RNA and DNA |
-
- 198.** Which of the following is not correct about viruses? (1)
- A.** All are parasites.
 - B.** They have different structure and shape.
 - C.** They have ability to synthesize nucleic acid and proteins.
 - D.** Antibiotics have no effect on them.
-
- 199.** Which of the following is not correctly paired with its function? (1)
- A.** Saprophytes - decomposition of dead organic material
 - B.** Parasites - Living on living plant or animal
 - C.** Lichens - Symbiotic association of plant and algae
 - D.** Mycorrhiza - symbiotic association of fungi with plant roots
-
- 200.** Which of the following is wrong about fungi? (1)
- A.** They are eukaryotic.
 - B.** All fungi possess cellulose cell wall.
 - C.** They are heterotrophic.
 - D.** They are both unicellular and multicellular.
-
- 201.** The cell wall of most fungi is composed of: (1)
- | | |
|-------------------------|--------------------------|
| A. Cellulose | B. Chitin |
| C. Peptidoglycan | D. Polysaccharide |
-
- 202.** Single celled eukaryotes are included in Kingdom (1)
- | | |
|--------------------|-------------------|
| A. Protista | B. Fungi |
| C. Monera | D. Archaea |
-
- 203.** Which of the following organism lack membrane organelle? (1)
- | | |
|-------------------------|-------------------------|
| A. Streptococcus | B. Chlamydomonas |
| C. Plasmodium | D. Saccharomyces |
-
- 204.** Lichens are organism produced by intimate association of (1)
- | | |
|------------------------------|----------------------------|
| A. algae and plant | B. fungus and algae |
| C. algae and bacteria | D. fungus and plant |
-
- 205.** Which of the following statement about viruses is correct? (1)
- A.** Viruses possess their own metabolic system
 - B.** Viruses are obligate parasites
 - C.** All viruses contain both RNA and DNA
 - D.** Nucleic acid of viruses is known as capsid

-
- 206.** Which of the following organism often used as indicator of pollution and sensitive to SO₂ (1)
- A. Lichens
B. Fungi
C. Mosses
D. Mushrooms
-
- 207.** Which of the following groups are decomposers of plant (1)
- A. Monera and Fungi
B. Protista and Animalia
C. Fungi and plants
D. Animalia and Monera
-
- 208.** Which one of the following is correctly paired with feature? (1)
- A. Nematoda → Animal with radial body symmetry
B. Arthropoda → soft bodied animals
C. Mollusca → spiny skinned animals
D. Sponges → hollow filter feeder animals
-
- 209.** Which of the following fungi does not cause Tinea capitis? (1)
- A. Trichophyton tonsurans
B. Microsporum audouinii
C. Trichophyton violaceum
D. Trichophyton rubrum
-
- 210.** Which of the following ringworm is also called Tinea barbae? (1)
- A. Ringworm of Scalp
B. Ringworm of Beard
C. Ringworm of Body
D. Ringworm of Groin
-
- 211.** Which virus has complex symmetry? (1)
- A. T4 bacteriophage
B. Influenza virus
C. Adenovirus
D. Papilloma viruses
-
- 212.** Which of the following is a helical virus? (1)
- A. Tobacco Mosaic Virus (TMV)
B. T4 bacteriophage
C. Herpesvirus
D. Poxvirus
-
- 213.** Which of the following statements is correct about Koch's postulate? (1)
- A. The cultured microbes should not cause disease when introduced to a healthy organism
B. Microorganism that causes diseases must be identified from the healthy individual.
C. The microorganism should be isolated from experimental host and are identical to the original type.
D. All microbes that cause disease are cultured in artificial medium.
-
- 214.** Which of the following disease is not transmitted by animal vector? (1)
- A. Elephantiasis
B. Trypanosomiasis
C. Leishmaniasis
D. Trichomoniasis
-
- 215.** Which of the following fungal species is responsible for the production of aflatoxin? (1)
- A. Aspergillus fumigatus
B. Claviceps purpurea
C. Aspergillus flavus
D. Tinea unguium

-
- 216.** Which of the following is asexual and sexual method of reproduction in ciliated protozoan respectively? (1)
- A. Transverse fission and conjugation
 - B. Longitudinal cell division and binary fission
 - C. Conjugation and Budding
 - D. Multiple fission and Cyst formation
-
- 217.** Which of the following genera of fungi is not responsible for causing ringworms? (1)
- A. Trichophyton
 - B. Microsporum
 - C. Fusarium
 - D. Epidermophyton
-
- 218.** Which of the following is not the ways in which protozoan obtain food? (1)
- A. Ingestion
 - B. Absorption
 - C. Phagocytosis
 - D. Exocytosis
-
- 219.** Which of the following is the feature of all organism in kingdom Monera? They are all. (1)
- A. Multicellular
 - B. Unicellular
 - C. Autotrophic
 - D. Parasitic
-
- 220.** Which of the following is not the structure of prokaryotic cell? (1)
- A. Ribosome
 - B. Slime capsule
 - C. Nucleoid
 - D. Nucleolus
-
- 221.** The main constituent of a Gram-positive cell wall is (1)
- A. Peptidoglycan
 - B. Glycocalyx
 - C. Flagella
 - D. Fimbriae
-
- 222.** Which of the following is classified as cell envelope in prokaryotic organism? (1)
- A. Capsomere
 - B. Cell inclusion
 - C. Cell membrane
 - D. Appendage
-
- 223.** Which of the following statement about bacterial structure is not correct? (1)
- A. The flagella allow substances in and out of the bacteria
 - B. Plasmid is a piece of extra chromosomal DNA
 - C. Bacteria are best measured in micrometer
 - D. Bacteria are omnipresent organism
-
- 224.** The Iodine used in gram staining serves as (1)
- A. Mordant
 - B. Cofactor
 - C. Decolorizer
 - D. Fixer
-
- 225.** Which of the following is acellular? (1)
- A. Archaea
 - B. Viroid
 - C. Yeast
 - D. Bacteria

-
- 226.** Which of the following prokaryotic structure is not correctly paired with its function? (1)
- A. Mesosome → site where large DNA molecules is condensed
 - B. Flagellum → specialized appendage that provides motility
 - C. Glycocalyx → coating layer that provide adhesive and receptor function
 - D. Pilus → hollow appendage used in transfer of DNA to another cell
-
- 227.** Which of the following is NOT the distinct feature of prokaryotes? (1)
- A. They lack membrane enclosed genetic material
 - B. Their cell is chemically simple
 - C. They lack histone associated protein
 - D. They often reproduce by binary fission
-
- 228.** Which of the following is not true about the cell wall of gram-positive bacteria? (1)
- A. Cell wall comprises of many layers
 - B. The cell wall is thicker than the associated gram-negative bacteria
 - C. Cell wall comprises of teichoic acids
 - D. Cell wall consists of thin layer of peptidoglycan
-
- 229.** An organism that uses methane as carbon source to derive energy is termed as (1)
- A. Chemolithoautotroph
 - B. Photoheterotroph
 - C. Methanotroph
 - D. Photolithoautotroph
-
- 230.** Chemo organoheterotrophs are organisms that (1)
- A. harvest energy from inorganic chemicals
 - B. use reduced organic compounds as source of energy
 - C. oxidize reduced inorganic compounds to derive both energy and electron for biosynthesis
 - D. obtain energy from sunlight and fix carbon dioxide
-
- 231.** Which of the following disease cannot be transmitted through droplet contact? (1)
- A. Influenza
 - B. Common cold
 - C. AIDS
 - D. Ebola
-
- 232.** Which of the following is a bacterial disease? (1)
- A. Leprosy
 - B. Amoebiasis
 - C. Measles
 - D. Thrush
-
- 233.** Nitrites are oxidized to nitrates by a microorganism called (1)
- A. Nitrosococcus
 - B. Azotobacter
 - C. Nitrobacter
 - D. Nitrosomonas
-
- 234.** Bacteria which strictly need oxygen for growth are called (1)
- A. obligate aerobes
 - B. facultative aerobes
 - C. obligate anaerobes
 - D. microaerophilic bacteria
-
- 235.** Bacteria reproduce vegetatively by (1)
- A. binary fission only
 - B. fragmentation and budding only
 - C. transformation and transduction
 - D. fission, fragmentation and budding

236. Binary fission in bacteria involves all except

(1)

- | | |
|-----------------------------------|---------------------------|
| A. Spindle fiber formation | B. DNA replication |
| C. Cytokinesis | D. Cell elongation |
-

237. The transfer of genetic material from one bacterium to another via virus is termed as

(1)

- | | |
|--------------------------|------------------------|
| A. transformation | B. transduction |
| C. conjugation | D. transfection |

— END OF EXAMINATION —

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